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S1-26_SEZG628T : Dissertation.

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DISSERTATION / PROJECT / PROJECT WORK TITLE	Design and Development of a Lightweight Runtime Quality Assessment Framework for Cloud-Native Microservice Applications

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**Design and Development of a Lightweight Runtime Quality Assessment Framework for
Cloud-Native Microservice Applications**

Course No. : (S1-26_SEZG628T)

Course Title : Dissertation / Project Work

Dissertation Done by :

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BITS ID:2024tm93686

Degree Program : M.Tech in Software Engineering

Research Area : Software Architecture, Runtime Quality

Assessment of Cloud-Native Microservices

Dissertation / Project Work carried out at :

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**BIRLA INSTITUTE OF TECHNOLOGY & SCIENCE, PILANI
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1. Broad Area of Work

Cloud-native software engineering and microservice architectures form the broad area of this dissertation. Modern applications are increasingly decomposed into independently deployable microservices running on containerized, orchestrated infrastructure. Ensuring and continuously assessing the runtime quality of such distributed systems is a growing research concern. The current work focuses on the following related sub-areas:

- Cloud-native software engineering and microservice architecture
- Observability, monitoring and telemetry of distributed systems (metrics, logs, traces)
- Software quality assessment, metrics engineering and composite indexing
- Multi-criteria decision-making for weight assignment (Analytic Hierarchy Process, Entropy Weight Method)
- DevOps and Site Reliability Engineering practices for cloud-native systems

2. Background

Organizations are rapidly re-architecting monolithic applications into cloud-native microservices to gain agility, scalability and independent deployability. This shift is directly relevant to the current work environment, where distributed services are monitored using tools such as Prometheus and Grafana that continuously generate large volumes of runtime metrics, including CPU usage, memory usage, response time, error rate, availability, throughput and network latency.

While these tools are effective at collecting metrics and rendering dashboards, they generally present each metric in isolation. Developers and operations teams are left to manually correlate multiple dashboards to judge whether the overall system is healthy, whether quality is improving or degrading across releases, and which microservice is contributing most to quality degradation. Two deployments or two versions of the same service cannot be objectively compared using a single number today. Existing literature on microservice architecture largely emphasizes monitoring, observability, logging and distributed tracing, or design-time metrics such as coupling and cohesion, but there is limited work on a runtime quality evaluation framework that combines multiple quality attributes, assigns them principled weights, and produces a single, interpretable Quality Index for a running microservice system. This gap motivates the proposed research.

3. Objectives

The objectives of this dissertation are as follows:

1. To study the runtime quality attributes relevant to cloud-native microservice applications.
2. To identify and select measurable runtime metrics collected by standard monitoring tools such as Prometheus.
3. To design a lightweight Quality Assessment Framework (QAF) comprising metric collection, preprocessing, quality analysis and indexing layers.
4. To develop a Quality Index (QI) calculation model that aggregates Performance, Reliability, Availability, Resource Efficiency and Scalability scores using a defined weighting scheme.
5. To implement the proposed framework on a representative microservice prototype (e.g., User, Product, Order and Payment services).
6. To validate the framework experimentally under varying workloads (low, medium, peak and failure-injected traffic) using load-testing tools.
7. To compare the results obtained from the proposed framework with traditional dashboard-based monitoring approaches.

4. Scope of Work

The scope of this dissertation, to be carried out independently by the student, is to design and develop a lightweight, tool-agnostic Quality Assessment Framework that converts multiple runtime metrics of a cloud-native microservice application into a single, meaningful Quality Index. The work includes:

- Defining quality dimensions — Performance, Reliability, Availability, Resource Efficiency and Scalability — and mapping each to measurable runtime metrics.
- Designing the Quality Index formula $QI = w1P + w2R + w3A + w4E + w5S$ and determining weights either through a fixed weighting scheme or through the Analytic Hierarchy Process / Entropy Weight Method.
- Implementing the framework modules: Metric Collector, Metric Preprocessing (normalization, noise removal, handling missing data), Quality Analyzer, Quality Index Calculator, Quality Classification Engine (Excellent / Good / Fair / Poor / Critical) and a basic Recommendation Engine.

- Building a proof-of-concept microservice application (Spring Boot or Node.js, containerized with Docker, orchestrated with Kubernetes or Docker Compose) instrumented with Prometheus and visualized through Grafana.
- Conducting load testing with JMeter or k6 under multiple workload and failure scenarios and validating the resulting Quality Index against observed system behaviour.

Aspects such as building a production-grade commercial monitoring product, large-scale multi-cluster deployment, and security/compliance auditing of the microservices are outside the scope of this dissertation.

5. Plan of Work

The following is the tentative plan of work for the semester:

Phase	Duration 2026	Work to be Done
Literature Review & Dissertation Outline	1 Aug – 15 Aug 2026.	Study of runtime quality attributes, observability literature, QoS models; finalize research problem, gap and outline.
Framework & Quality Index Model Design	16 Aug – 30 Aug 2026.	Design of the Quality Assessment Framework (QAF); definition of quality dimensions (Performance, Reliability, Availability, Resource Efficiency, Scalability); formulation of the Quality Index (QI) model and weighting scheme (AHP / Entropy Weight Method).
Prototype Development	01 Sep - 15 Sep 2026.	Development of a sample microservice application (User, Product, Order, Payment services) using Docker/Kubernetes; integration of Prometheus & Grafana; implementation of Metric Collector, Preprocessing, Quality Analyzer and QI Calculator modules.
Experimentation & Validation	15 Sep – 04 Oct 2026.	Load testing using JMeter/k6 under low, medium, peak and failure-injected

		workloads; computation of QI; comparison against traditional dashboard-based monitoring.
Dissertation Outline Review	5 Oct -19 Oct 2026.	Submission of outline to Supervisor & Additional Examiner for review and feedback.
Final Compilation & Submission	20 Oct - 5 Nov 2026.	Incorporation of feedback, final documentation and submission of the outline report.

6. Literature References

The following is an indicative list of references identified during the preliminary literature review; the list will be expanded as the literature survey progresses:

1. V. C. Tapia et al., "Research Opportunities in Microservices Quality Assessment: A Systematic Literature Review," *Journal of Advances in Information Technology*, vol. 14, no. 5, 2023.
2. "A Metrics Framework for Evaluating Microservices Architecture Designs," *IEEE / River Publishers Journals*, 2023 — proposes Service Granularity, Lack of Cohesion and Number of Operations metrics for microservice quality evaluation.
3. "Automated Assessment of the Relationship between Microservice Architectures and Performance," *Journal of Systems and Software (ScienceDirect)*, 2026 — combines static architecture analysis with dynamic performance testing.
4. "Fostering Microservice Maintainability Assurance through a Comprehensive Framework," *arXiv:2407.16873*, 2024 — catalogues microservice anti-patterns and runtime assessment metrics.
5. "A Decomposition and Metric-Based Evaluation Framework for Microservices," *arXiv:1908.08513* — metric-based evaluation of microservice decomposition quality.

6. *"A Multidimensional Decision-Support Framework for Software Quality Assessment in Agile Projects (OSQI)," Information (MDPI), 2026 — integrates AHP and entropy-based weighting into a composite software quality index.*
7. *T. L. Saaty, "The Analytic Hierarchy Process: Planning, Priority Setting, Resource Allocation," McGraw-Hill, 1980 — foundational work on AHP for multi-criteria weight assignment.*
8. *"Systematic Mapping Study of Test Generation for Microservices: Approaches, Challenges, and Impact on System Quality," Electronics (MDPI), 2025.*

8. Particulars of the Supervisor and Additional Examiner

	Supervisor	Additional Examiner
Name	Ajay Singh	MASAWWAR ALI
Qualification	B.Tech	MCA
Designation	Lead Architecture	Teach Lead
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9. Remarks of the Supervisor

- The proposed topic addresses a genuine gap in current microservice monitoring practice — moving from isolated dashboard metrics to a single, interpretable Quality Index — and is well suited to an M.Tech R&D dissertation.
- The problem statement and research gap are clearly articulated, with adequate justification drawn from the observability and software-quality-assessment literature.
- The objectives are specific, measurable and sequenced logically, progressing from literature study to framework design, implementation and experimental validation.
- The scope of work is realistic for a single-semester outline and appropriately distinguishes framework design (the research contribution) from the prototype (a proof of concept).
- The proposed use of AHP / Entropy Weight Method for determining quality-dimension weights adds methodological rigour beyond simple manual weight assignment.
- The plan of work allocates sufficient time to experimentation under varied workloads, which is essential for validating the Quality Index empirically rather than only theoretically.
- The preliminary literature review is adequate at this stage; the student is advised to expand it with recent peer-reviewed work on QoS-based composite indices and to benchmark against at least one existing monitoring approach.
- Based on the above, the topic and outline are approved, and the student is permitted to proceed with the dissertation as planned, subject to periodic review of progress.

Information about the Supervisor:

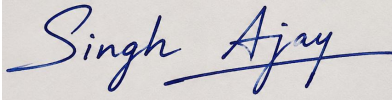
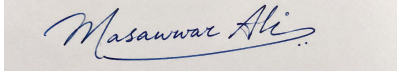
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